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SUPPLEMENT TO "ABS CHALLENGES: WINS LEFT ON THE TABLE": PROOFS, SYMBOLIC VERIFICATION AND ADDITIONAL RESULTS

BY JAN JANOUŠEK[1] AND MICHAL PEŠTA[1,a]

[1]*Department of Probability and Mathematical Statistics, Faculty of Mathematics and Physics, Charles University, Prague, Czech Republic,* [a]*pesta@karlin.mff.cuni.cz*

Section S1 proves Theorems 2–4 of the main text and a comparison of retention with consumable challenges, and marks which statements are proved and which are numerical only. Section S2 lists the Mathematica checks behind the paper's algebra. Section S3 is the geometry and data audit. Section S4 gives the perception model in full: every variant, player heterogeneity, out-of-sample and minor-league fits, identification and the sensitivity of the wins lost to perception noise. Section S5 documents the policy evaluation: the causal re-simulation, the competitor rules, the decomposition of the gap between the simulator and the WP engine, the flat-optimum bound on the calibrated pools and the bootstrap. Section S6 gives the 2026 postseason check in full, and Section S7 describes the cricket DRS replication.

Equation, theorem and section numbers without the prefix S refer to the main text.

S1. Proofs. Each statement carries a status: [PROVED] means the proof is complete here and its algebraic steps are checked in Mathematica (Section S2); [PROVED, SKETCH] means the argument is complete but uses a standard lemma that is cited rather than reproduced; [NUMERICAL] means the statement rests on deterministic computation only. Table S2 at the end of the section collects them.

S1.1. *Setting.* Let t be the number of opportunities left and c the challenges held. At each opportunity a pair (p, g) with law F_t arrives: $g \geq 0$ is the stake and $p \in [0,1]$ the calibrated probability that a challenge succeeds. A successful challenge gains g and keeps the challenge; a failed one loses it. With $V_0(\cdot) = V_t(0) = 0$ the value of challenging at (t, c) is $p\{g + V_{t-1}(c)\} + (1-p)V_{t-1}(c-1) = V_{t-1}(c) + pg - (1-p)\Delta_{t-1}(c)$ and the value of waiting is $V_{t-1}(c)$, which gives the Bellman equation (5),

$$V_t(c) = V_{t-1}(c) + \varphi_t\{\Delta_{t-1}(c)\}, \qquad \varphi_t(d) = \mathbb{E}_t\{(pg - (1-p)d)^+\},$$

and the cutoff (6): challenge iff $pg \geq (1-p)\Delta_{t-1}(c)$, that is $p \geq \Delta_{t-1}(c)/\{g + \Delta_{t-1}(c)\}$. The Q-gap between the two actions is

$$|pg - (1-p)D| = |p(g+D) - D|, \qquad D = \Delta_{t-1}(c). \tag{S1}$$

Throughout, $\psi_t(d) = d + \varphi_t(d)$. Since $V_t(0) = 0$, $\Delta_t(1) = V_t(1)$, and the Bellman equation gives

$$\Delta_t(1) = \psi_t\{\Delta_{t-1}(1)\}, \qquad \Delta_t(c) = \psi_t\{\Delta_{t-1}(c)\} - \varphi_t\{\Delta_{t-1}(c-1)\} \quad (c \geq 2). \tag{S2}$$

S1.2. *Monotone thresholds.*

LEMMA S1 (PROVED). *For any law F of (p, g) on $[0,1] \times [0, \infty)$ with $\mathbb{E}g < \infty$, $\varphi(d) = \mathbb{E}\{(pg - (1-p)d)^+\}$ is nonnegative and nonincreasing on $[0, \infty)$, with difference quotients in $[-1, 0]$, and $\psi(d) = d + \varphi(d)$ is nondecreasing.*

PROOF. Fix (p, g). The map $j(x) = (pg - (1-p)x)^+$ is the positive part of an affine function of slope $-(1-p) \in [-1, 0]$, so it is nonnegative, nonincreasing, and its difference quotients lie in $[-(1-p), 0] \subset [-1, 0]$. Then $x + j(x)$ has difference quotients in $[0, 1]$. Taking expectations (finite because $j(x) \le g$) preserves all three properties. □

Theorem 2 (main text). *Under Assumption 1, for all t and $c \ge 1$: (a) $\Delta_t(c) \ge \Delta_t(c+1) \ge 0$; (b) $\Delta_t(c) \ge \Delta_{t-1}(c)$.*

PROOF [PROVED]. By induction on t. For $t = 0$, $\Delta_0 \equiv 0$ and both claims hold. Suppose they hold at $t-1$ and fix $c \ge 1$. Write $a = \Delta_{t-1}(c-1)$ (only for $c \ge 2$), $b = \Delta_{t-1}(c)$ and $e = \Delta_{t-1}(c+1)$; by hypothesis $a \ge b \ge e \ge 0$. Under Assumption 1 the same function φ_t applies at every c, so by (S2)

$$\Delta_t(c) = \psi_t(b) - \varphi_t(a), \qquad \Delta_t(c+1) = \psi_t(e) - \varphi_t(b),$$

with the $\varphi_t(a)$ term absent when $c = 1$. By Lemma S1, $\psi_t(b) \ge \psi_t(e)$ and $\varphi_t(b) \ge \varphi_t(a)$, so $\Delta_t(c) - \Delta_t(c+1) = \{\psi_t(b) - \psi_t(e)\} + \{\varphi_t(b) - \varphi_t(a)\} \ge 0$, which is (a). Next $\Delta_t(c) - \Delta_{t-1}(c) = \varphi_t(b) - \varphi_t(a) \ge 0$ (and $= \varphi_t(b) \ge 0$ for $c = 1$), which is (b); with $b \ge 0$ it also gives $\Delta_t(c) \ge 0$. The optimal cutoff $D/(g+D)$ is increasing in D (its derivative in D is $g/(g+D)^2 > 0$), so it falls as t decreases and rises as c decreases. □

REMARK S1 (Where the theorem fails). *A law that depends on c.* Let $g = 1$, $p = 1/10$ when one challenge is held and $p = 9/10$ when two are held. After five opportunities the exact DP gives $\Delta(1) = 0.111$ and $\Delta(2) = 3.61$, so V is convex in c: the second challenge is worth more than the first because holding it changes the opportunities themselves. Exogeneity in Assumption 1 cannot be dropped. *The extra-inning top-up.* The calibrated DP refills $c = 0$ to $c = 1$ at the start of each extra inning, which sets $V(0) = V(1)$ at the last regulation opportunity of the home side. There, and only there, $\Delta(1) = 0 < \Delta(2)$; every other row of the calibrated DP satisfies both monotonicities on both sides. *Exact random laws.* On 300 seeded random discrete laws (i.i.d. and time-varying, p and g dependent, retention and consumption) the exact rational DP satisfies both monotonicities in every case.

Relation to the stochastic knapsack. In Derman, Lieberman and Ross (1972) and Kleywegt and Papastavrou (1998) a unit of capacity is spent whenever an item is accepted, and the analogue of φ is $\mathbb{E}(pg - d)^+$. Under retention the unit is spent only on failure, so the continuation value enters the gain as $p(g + \Delta)$. The induction above uses only the properties in Lemma S1, which both gain functions share.

S1.3. *The square-root law.* **Theorem 3 (main text).** *Let $c = 1$, $F_t = F$ for all t, $q = 1 - p$, and suppose that given g, q has a density $f(\cdot \mid g)$ near 0 with $f(0+ \mid g) = h(g)$, bounded by $M(g)$ on $[0, x_0]$ with $\mathbb{E}[g^2 M(g)] < \infty$, $\mathbb{E}g^2 < \infty$, and $K = \mathbb{E}[h(g)g^2]/2 \in (0, \infty)$. Then $\Delta_t(1) \sim \sqrt{\mathbb{E}[h(g)g^2]\, t}$.*

PROOF [PROVED]. Write $\Delta_t = \Delta_t(1)$ and $\varphi = \varphi_t$.

Step 1: $\Delta_t \to \infty$. By Theorem 2(b), Δ_t is nondecreasing. If $\Delta_t \le L$ for all t, then by (S2) and Lemma S1, $\Delta_t = \Delta_{t-1} + \varphi(\Delta_{t-1}) \ge \Delta_{t-1} + \varphi(L)$, so $\Delta_t \ge t\varphi(L) \to \infty$, a contradiction as soon as $\varphi(L) > 0$. And $\varphi(L) = \mathbb{E}\{(g - q(g+L))^+\} > 0$ because $K > 0$ forces $h(g)g^2 > 0$

on a set of positive probability, on which q has positive density near 0, so that $q < g/(g+L)$ with positive probability.

Step 2: $d\varphi(d) \to K$ as $d \to \infty$. Since $pg-(1-p)d = g-q(g+d)$, the integrand is positive iff $q < g/(g+d)$. Let $c_0 = x_0/(1-x_0)$ and split $d\varphi(d) = A(d) + B(d)$ over $\{g \le c_0 d\}$ and $\{g > c_0 d\}$. On the first set $g/(g+d) \le x_0$, so the positive part only involves $q \in [0, x_0]$, where q has the density f, and substituting $q = ug/(g+d)$ (the integrand becomes $g(1-u)$ and $dq = g\,du/(g+d)$) gives

$$A(d) = \mathbb{E}\Big[\frac{dg^2}{g+d}\int_0^1 (1-u)\, f\Big(\frac{ug}{g+d}\,\Big|\, g\Big)du\,;\, g \le c_0 d\Big].$$

For fixed g and u the integrand converges to $g^2(1-u)h(g)$, whose expectation is K, and it is at most $g^2 M(g)/2$, so $A(d) \to K$ by dominated convergence. On the second set the density is not used: $B(d) = d\,\mathbb{E}[(g-q(g+d))^+;\, g > c_0 d] \le \mathbb{E}[dg;\, g > c_0 d] \le \mathbb{E}[g^2 \mathbf{1}\{g > c_0 d\}]/c_0 \to 0$. Hence $d\varphi(d) \to K$.

Step 3. By (S2), $\Delta_t^2 - \Delta_{t-1}^2 = \eta(\Delta_{t-1})$ with $\eta(d) = 2d\varphi(d) + \varphi(d)^2$ (identity (7)). By Steps 1–2, $\varphi(\Delta_t) \to 0$ and $\eta(\Delta_t) \to 2K$, so by Cesàro $\Delta_t^2/t = t^{-1}\sum_{s \le t} \eta(\Delta_{s-1}) \to 2K$. □

Step 2 needs $\mathbb{E}g^2 < \infty$ to control the large-stake set. It holds trivially for stakes in win probability ($0 \le g \le 1$), and the statement in the main text includes it.

REMARK S2 (Other rates, [PROVED, SKETCH]; constants exact). If $\mathbb{P}(q \le x \mid g) \sim c(g)x^\alpha$, the same substitution gives $d^\alpha \varphi(d) \to K_\alpha = \mathbb{E}[c(g)g^{1+\alpha}]/(1+\alpha)$. By the mean value theorem $\Delta_t^{1+\alpha} - \Delta_{t-1}^{1+\alpha} = (1+\alpha)\xi^\alpha \varphi(\Delta_{t-1})$ with $\xi \in [\Delta_{t-1}, \Delta_t]$ and $\xi/\Delta_{t-1} \to 1$, so the increments converge to $(1+\alpha)K_\alpha$ and $\Delta_t \sim \{(1+\alpha)K_\alpha t\}^{1/(1+\alpha)}$. The domination step for $\alpha \ne 1$ is analogous to Step 2 and is not written out; it needs $\mathbb{E}[c(g)g^{1+\alpha}] < \infty$ and a local bound on $\mathbb{P}(q \le x \mid g)/x^\alpha$ that is integrable against $g^{1+\alpha}$, the analogues of the conditions of Theorem 3. Numerically, at $t = 10^6$: with a positive finite density of p at 1 ($\alpha = 1$, $K_1 = 9/8$ exactly) $\Delta_t/\sqrt{2K_1 t} = 0.9986$; with a density vanishing linearly at 1 ($\alpha = 2$, $K_2 = 83/144$) $\Delta_t/(3K_2 t)^{1/3} = 0.985$; with an atom at $p = 1$ the growth is linear, $\Delta_t/t = 0.0917$ against the limit $\mathbb{E}[g; p = 1] = 11/120 = 0.0917$.

REMARK S3 (Time-varying laws, [PROVED, SKETCH]). If F_t varies with t, $K_t \in [K_{\min}, K_{\max}] \subset (0, \infty)$ and Step 2 holds uniformly in t, the same argument with weighted Cesàro sums gives $\Delta_t^2/\sum_{s \le t} 2K_s \to 1$: the option value is the square root of the cumulative squared-stake mass. With stakes alternating between 1 and 3 and uniform p, the ratio is 0.9973 at $t = 10^6$.

REMARK S4 (More challenges, [PROVED, SKETCH]; constants exact). For $c \ge 2$, $\Delta_t(c) \sim k_c\sqrt{2Kt}$ with $k_1 = 1$ and $k_c^2 + k_c/k_{c-1} = 1$. Sketch: $\Delta_t(c) \to \infty$ by the argument of Step 1; $y_t = \Delta_t(c)^2/(2Kt)$ satisfies a deterministic stochastic-approximation recursion with step $1/t$ and drift $1 - \sqrt{y}/k_{c-1} - y + o(1)$, which is strictly decreasing with unique root k_c^2; the Robbins–Monro/Chung lemma (e.g. Kushner and Yin, 2003, Ch. 5) gives $y_t \to k_c^2$: the recursion is deterministic, its steps are $O(1/t)$, the drift is continuous and strictly decreasing with a unique root, and the $o(1)$ term comes from $\Delta_t(c-1)/\{k_{c-1}\sqrt{2Kt}\} \to 1$, the induction hypothesis; a full proof would write out that error bound. Hence $k_2 = (\sqrt{5}-1)/2$, and $k_1, \ldots, k_6 = 1.000, 0.618, 0.477, 0.401, 0.351, 0.316$. With uniform p and $g = 1$, $\Delta_t(c)/\sqrt{t}$ for $c = 1, \ldots, 4$ at $t = 10^6$ is 0.9990, 0.6170, 0.4763, 0.3997.

REMARK S5 (Consumption, [PROVED]). Under consumption, $\Delta_t^{\mathrm{cons}}(c) \le s$ for all t, where $s = \operatorname{ess\,sup}(pg)$. The inductive step is $x \le s,\ Y \le s \Rightarrow x + \mathbb{E}(Y - x)^+ \le s$. With uniform p and $g = 1$, Δ^{cons} at $t = 10^6$ is 0.999998.

REMARK S6 (The calibrated game, [NUMERICAL]). In the calibrated 2026 DP, Δ_1 is of the same order as the stakes, so the game is pre-asymptotic. What carries over exactly is identity (7): for uniform p, $\eta(d) = g^2(2d+g)^2/\{4(d+g)^2\}$ rises only from $g^2/4$ to g^2 as d/g goes from 0 to ∞, so Δ^2 is close to linear in the opportunities left even far from the asymptote. The calibrated p is concentrated near zero, where η varies much more, so for the calibrated game the near-linearity below is an empirical finding rather than a consequence of the identity. In the score-aware DP the score is part of the state, so the law of (p, g) is no longer exogenous; the object that Theorem 3 speaks to is the continuation value averaged over the score states the 2026 games actually visit. For that average, a linear fit of Δ_1^2 on the expected regulation opportunities left n has $R^2 = 0.9971$ (home) and 0.9977 (away), and $\Delta_1 \approx a + 0.0020\sqrt{n}$ has $R^2 = 0.9955$ and 0.9991, against 0.98 for a linear fit in n. Along a path held at a tied score the fit is poor (R^2 of 0.85–0.88): late in a tied game the continuation value stays high, because every remaining call matters. We therefore say that the option value *scales like* $\sqrt{n}$ on average over game states, not that it follows the asymptotic law; an R^2 near one on a smooth monotone curve discriminates only weakly between exponents, and the linearity of Δ^2 is what identity (7) suggests. The ratio Δ_2/Δ_1 at the first pitch is 0.53–0.54, below the limit 0.618, as the pre-asymptotic iteration predicts.

S1.4. *The flat optimum.* **Theorem 4 (main text).** *(i) Exactly,*

$$V_T^\star(c_0) - V_T^\delta(c_0) = \sum_t \sum_{c \ge 1} \mathbb{P}^\delta(c_t = c)\, \mathbb{E}_g \int_{I_t(c,g)} |p(g+D) - D|\, dF_t(p \mid g).$$

(ii) If the conditional density of p on I_t is at most M, then $V_T^\star - V_T^\delta \le (M/2) \sum_t \sum_c \mathbb{P}^\delta(c_t = c)\, \mathbb{E}_g[g^2(D-\delta)^2/\{(g+D)(g+\delta)^2\}]$.

PROOF [PROVED]. (i) Let $Q_t^\star(c, a)$ be the value of taking action a at (t, c) and acting optimally afterwards, and V_t^δ the value of following ρ_δ. For the action $a = \rho_\delta(t, c, p, g)$,

$$V_t^\star(c) - V_t^\delta(c) = \{V_t^\star(c) - Q_t^\star(c, a)\} + \mathbb{E}\{V_{t-1}^\star(c') - V_{t-1}^\delta(c') \mid t, c, a\},$$

because $Q_t^\star(c, a)$ and the value of a under ρ_δ share the immediate reward and the transition to c' and differ only in the continuation. Unrolling from (T, c_0) along the trajectory of ρ_δ gives $V_T^\star(c_0) - V_T^\delta(c_0) = \mathbb{E}^\delta \sum_t \{V_t^\star(c_t) - Q_t^\star(c_t, a_t)\}$ (Kakade and Langford, 2002; Puterman, 1994). The bracket is zero when ρ_δ takes the optimal action and equals the Q-gap (S1) otherwise. With $c_t = c \ge 1$ the two rules differ exactly for p between $D/(g+D)$ and $\delta/(g+\delta)$, which is the interval $I_t(c, g)$. The Q-gap vanishes at the first endpoint. At the second, ρ_δ challenges when $p = \delta/(g+\delta)$ exactly (the convention of the policy code), so I_t is open there when $\delta < D$ and closed when $\delta > D$; with this convention (i) holds for any law of p, atoms included. The values compared are those before (p, g) is observed at t; the decomposition conditions on (p, g) inside the expectation. Finally, c_t is a function of the past and by Assumption 1 the current (p, g) is independent of it, so conditioning on $c_t = c$ leaves the law F_t unchanged. States with $c = 0$ have no action and contribute nothing.

(ii) On $I_t(c, g)$ the Q-gap is $(g+D)|p - D/(g+D)|$, linear in p and zero at the endpoint $D/(g+D)$. The interval has length $L = g|D-\delta|/\{(g+D)(g+\delta)\}$. With density at most

M, $\int_I (g+D)|p - D/(g+D)|\,dF \le M(g+D)\int_0^L u\,du = M(g+D)L^2/2 = Mg^2(D-\delta)^2/\{2(g+D)(g+\delta)^2\}$.

(iii, crude form) Since $g+D \ge g$ and $(g+\delta)^2 \ge 4g\delta$, $g^2/\{(g+D)(g+\delta)^2\} \le 1/(4\delta)$, so the regret is at most $(M/8\delta)\sum_t \mathbb{E}^\delta[(\Delta_{t-1}(c_t)-\delta)^2; c_t \ge 1]$. □

Envelope. The unit-density integral underlying bound (ii), $r(\delta) = g^2(D-\delta)^2/\{2(g+D)(g+\delta)^2\}$ (the integral of the Q-gap over I_t against dp), has $r(D) = r'(D) = 0$ and $r''(D) = g^2/(g+D)^3$. It is not the per-state regret of a general law of p (with $g = D = 1$, $\delta = 2$ and density $2p$ the regret is $11/324$, $r = 1/36$; wolfram/regret.wl), but it bounds that regret by $M\,r(\delta)$: under a bounded density, mis-setting the constant costs nothing to first order, so the loss of the best constant is at most quadratic in the spread of Δ along the path. Where the best δ sits depends on the density near each cutoff and on how the occupation of states moves with δ; it is near a weighted mean of Δ only when those weights are roughly constant. For a law with atoms, (ii) with a histogram height in place of M is an approximation, not a bound (Section S5.7).

Stylized illustration [NUMERICAL]. In a stylized game with 60 opportunities, two challenges, stakes rising through the game and a mostly low posterior (p density $0.9\,\mathrm{Beta}(1,4) + 0.1\,\mathrm{Beta}(4,1)$), Δ ranges over $[0, 0.065]$ along the path. The best constant $\delta = 0.0245$ loses 7.6% of the optimum; identity (i) holds to 4×10^{-17}; bound (ii) gives 15.3% and the crude form (iii) 185%, so (ii) is the form to use. The loss curve is flat near its minimum: 35 % at $\delta = 0.01$, 9 % at $\delta = 0.02$, 9 % at $\delta = 0.03$, 23 % at $\delta = 0.05$. The theorem on the calibrated 2026 pools is in Section S5.7.

S1.5. *Retention against consumable challenges.* Three allotment rules: *retention* (MLB ABS, a success keeps the challenge); *consumption* (every challenge is spent); and the *NBA cap* (one challenge; a success earns a second, consumable one, so at most two are used). Let $s = \operatorname{ess\,sup}(pg)$.

THEOREM S2 (PROVED). *Under Assumption 1: (a)* $V_t^{\mathrm{cons}}(c) \le V_t^{\mathrm{ret}}(c)$ *and*

$$V_t^{\mathrm{cons}}(1) \le V_t^{\mathrm{NBA}} \le \min\{V_t^{\mathrm{ret}}(1), V_t^{\mathrm{cons}}(2)\};$$

(b) $V_t^{\mathrm{cons}}(c) \le cs$ *and* $V_t^{\mathrm{NBA}} \le 2s$ *for all* t*; (c) if* $F_t = F$ *and there is* $g_0 > 0$ *with* $\mathbb{P}(p > 1-\varepsilon,\, g > g_0) > 0$ *for every* $\varepsilon > 0$*, then* $V_t^{\mathrm{ret}}(1) \to \infty$*, and under the conditions of Theorem 3,* $V_t^{\mathrm{ret}}(1) \sim \sqrt{\mathbb{E}[h(g)g^2]\,t}$*. Hence one retained challenge eventually beats any fixed allotment of consumable ones.*

PROOF. (a) By coupling. Run both rules on the same sequence of (p, g) and the same success events. The player under the richer rule copies the poorer player's actions and, wherever the poorer rule has removed a challenge that the richer rule keeps, stops challenging. This is feasible because on every path the richer rule holds at least as many challenges. NBA against consumption with one: after the single use consumption holds none. Retention with one against NBA: after a success both hold one, after a failure both hold none, and once NBA's consumable second challenge is spent retention may still hold one. Consumption with two against NBA: after a success both hold one; after a failure NBA holds none and consumption one. Retention against consumption with the same c: retention never holds fewer. Along coupled paths the gains are identical, so the richer rule's optimal value is at least the poorer one's. (b) At most c (respectively two) challenges are used, and each use has conditional expected gain $pg \le s$. (c) Fix $L \ge 0$ and let $\varepsilon = g_0/\{2(g_0+L)\}$. On $\{q \le \varepsilon,\ g > g_0\}$, with $q = 1-p$, $pg - (1-p)L = g - q(g+L) \ge g(1-\varepsilon) - \varepsilon L \ge g_0 - \varepsilon(g_0+L) = g_0/2$, so $\varphi(L) \ge (g_0/2)\mathbb{P}(p \ge 1-\varepsilon,\, g > g_0) > 0$. Step 1 of the proof of Theorem 3 then applies, and

the rate is that theorem. With (b), $V_t^{\mathrm{ret}}(1) > cs \geq V_t^{\mathrm{cons}}(c)$ for all large t. The stake bound g_0 cannot be dropped: with $q = g = U \sim \mathrm{Unif}(0,1)$, $\mathbb{P}(p > 1-\varepsilon,\, g > 0) = \varepsilon > 0$ for every ε, but $\varphi(L) = \mathbb{E}\{U(1-U-L)^+\} = (1-L)^3/6$ vanishes for $L \geq 1$, so $\Delta_t(1) < 1$ for all t ($\Delta_{2000}(1) = 0.961$). □

TABLE S1
Values under the three allotment rules, uniform p and $g = 1$ ($s = 1$), exact DP. The ordering of Theorem S2(a) holds at every t computed.

t	consumption, 1	NBA cap	retention, 1	consumption, 2	retention, 2
1	0.50	0.50	0.50	0.50	0.50
2	0.62	0.83	0.83	1.00	1.00
5	0.78	1.26	1.56	1.41	2.15
10	0.86	1.52	2.42	1.64	3.54
50	0.96	1.87	6.22	1.91	9.68
100	0.98	1.93	9.11	1.95	14.36
1,000	1.00	1.99	30.67	1.99	49.24
10,000	1.00	2.00	99.02	2.00	159.83

Table S1 gives the values in the uniform case: one retained challenge beats 2, 3 and 5 consumable ones for all $t \geq 4$, 10 and 27 respectively. In the stylized game above (60 opportunities, WP per team-game) the values are 0.032 (consumption, one), 0.048 (NBA cap), 0.055 (consumption, two), 0.066 (retention, one) and 0.098 (retention, two): two retained challenges are worth 77% more than two consumable ones, and a single retained challenge already beats two consumable ones.

TABLE S2
Status of the theoretical statements.

Statement	Status
Lemma S1; Theorem 2 (any exogenous, time-varying law)	proved
Failure for c-dependent laws; top-up row of the calibrated DP (Remark S1)	counterexample; numerical
Theorem 3, with its exact constant	proved
Rate $t^{1/(1+\alpha)}$ for $\alpha \neq 1$ (Remark S2)	proved, sketch; constants exact
Time-varying laws, $\Delta_t^2 \sim \sum_s 2K_s$	proved, sketch
$c \geq 2$: $k_c^2 + k_c/k_{c-1} = 1$, $k_2 = (\sqrt{5}-1)/2$ (Remark S4)	proved, sketch; constants exact
Consumption bound $\Delta^{\mathrm{cons}} \leq s$	proved
$\sqrt{n}$ scaling of the calibrated, score-averaged continuation value	numerical
Theorem 4: identity, bound and envelope	proved
Loss of the one-line rule on the calibrated pools and on the real stream (Section S5.7)	numerical
Theorem S2; crossover points	proved; numerical

S2. Symbolic verification. The exact algebra of the paper is checked in Mathematica (Section 8). All checks are deterministic: they use exact rationals, closed forms, `Reduce`/`Resolve` over the reals, or numerical integration at 30 digits, never sampling. The scripts:

- wolfram/check.wl → results/check.json: the retained-challenge cutoff and its monotonicity, and the Gaussian convolution used when tracking noise is added to perception noise.
- wolfram/identification_det.wl → results/identification-det-check.json: Proposition 1, identification of the deterministic-truth model (checks D1–D4).
- wolfram/identification.wl → results/identification-check.json: the alternative smooth-truth model, in which only the combined scale of tracking and perception noise enters the challenge curve. It is kept because it shows why a known truth matters (Section 3.4).
- wolfram/proofs.wl → results/proofs.json: the theory of Section S1 (T1–T4), including every numerical value quoted there.
- wolfram/supplement.wl → results/supplement-check.json: the algebraic steps added in this supplement (substitution and domination in the proof of Theorem 3, bound (ii), the mean-value step).
- wolfram/headline.wl → results/headline-check.json: the per-state regret, the causal decomposition of Section S5.3 and the cutoff form of the competitor rules.
- wolfram/drs_check.wl and wolfram/drs_audit.wl → results/drs/: the closed forms of the DRS model (Section S7).

Table S3 lists the 31 checks and their outcomes; 0 fail.

TABLE S3

Symbolic checks. "True": Mathematica returned `True` *for an identity or a quantified inequality. "Max. error": the largest absolute difference between a closed form and direct high-precision integration, or between the value computed in Python and the exact one.*

Script	Claim	Outcome
`check.wl`	Cutoff: challenge iff $p \geq \Delta/(g+\Delta)$ (no counterexample)	True
`check.wl`	Cutoff increasing in Δ, decreasing in g	True
`identification_det.wl`	D1: deterministic truth is the $\sigma_m \to 0$ limit of the smooth truth model	max. error 3×10^{-8}
`identification_det.wl`	D2: two-point Jacobian determinant $\phi_1\phi_2(w_1 - w_2)/\sigma_r^3$	0 (exact)
`identification_det.wl`	D2: rank of the challenge-curve Jacobian on the data grid, batter / catcher	2 / 2
`identification_det.wl`	D3: cross information of π with (σ_r, κ)	0 (exact)
`identification_det.wl`	D4: tracking noise c gives $\Phi\{(w-\kappa)/(\sigma_r^2 + c^2)^{1/2}\}$	max. error 2×10^{-25}
`check.wl`	Gaussian convolution of tracking and perception noise (symbolic and 30-digit)	True; max. error 4×10^{-29}
`identification.wl`	I1, I3: in the superseded smooth-truth model only $(\sigma_m^2 + \sigma_r^2)^{1/2}$ enters the challenge curve	0 (exact); rank 2 of 3
`proofs.wl`	T1a: Lemma S1 for any law (ForAll over the reals)	True
`proofs.wl`	T1b: induction step of Theorem 2	True
`proofs.wl`	T1c: both monotonicities in exact DPs on 300 random laws	True
`proofs.wl`	T1d: concavity fails for a law that depends on c	True
`proofs.wl`	T2a: $d\varphi(d) \to g^2 h/2$ and $d^\alpha \varphi(d) \to g^{1+\alpha}/(1+\alpha)$	True
`proofs.wl`	T2b: square-increment identity (7)	0 (exact)
`proofs.wl`	T2d: exact constants $K_1 = 9/8$, $K_2 = 83/144$	True
`proofs.wl`	T2e: $k_2 = (\sqrt{5}-1)/2$; SA drift decreasing with root k_c^2	True
`proofs.wl`	T2f: consumption step $x \leq s$, $Y \leq s \Rightarrow x + \mathbb{E}(Y-x)^+ \leq s$	True
`proofs.wl`	T3a: per-state regret on both sides of D; crude bound	True
`proofs.wl`	T3b: regret identity in the stylized game	gap 4×10^{-17}
`supplement.wl`	Substitution and domination bounds in the proof of Theorem 3	True
`supplement.wl`	Bound (ii) of Theorem 4; crude bound (iii)	True
`supplement.wl`	Mean-value step for the rate $t^{1/(1+\alpha)}$	True
`supplement.wl`	Theorem S2(c): bound $\varphi(L) \geq g_0/2 \cdot \mathbb{P}(\cdot)$; counterexample without g_0	True
`headline.wl`	Per-state regret, both sides; $r'(D) = 0$	True
`headline.wl`	Causal decomposition and RB difference (Section S5.3)	True
`headline.wl`	Runs rule and one-line rule are cutoff rules in p	True
`drs_check.wl`	DRS posterior and marginal; inversion of the review rule	True
`drs_check.wl`	DRS orthant correlation; bivariate normal CDF of the code	True; max. error 1×10^{-16}
`drs_check.wl`	DRS review rule with a retained umpire's call	True
`drs_audit.wl`	DRS three-way likelihood and posterior loss probability at the Test fit	max. error 7×10^{-17}

S3. Geometry and data audit.

S3.1. *Which plane each source reports.* Statcast publishes, for every pitch, a nine-parameter constant-acceleration trajectory and a location (`plate_x`, `plate_z`). Anchoring the trajectory on the reported release point, we find the distance y^* from the tip of the plate at which the trajectory passes through the reported location. Table S4 shows that through

2025 the location is at the front of the plate ($y = 17/12$ ft) and from 2026 at its midpoint ($y = 17/24$ ft), where it agrees with the trajectory to a few thousandths of an inch. The Stats API live feed (`pitchData.coordinates.pX/pZ`) stayed at the front of the plate. Its trajectory parameters and zone lines are Statcast's (identical on 100.0% and 99.96% of challenged pitches), and on the 10,544 challenged pitches its height is 0.95 in (SD 0.27) above Statcast's; projected to the midpoint the difference is 0.003 in. The Savant documentation says the same of `plate_x` and `plate_z`: "Through 2025, this was front-of-plate. From 2026 on, this is middle-of-plate to align with the ABS system" (Baseball Savant, 2026b).

TABLE S4

Plane of the reported pitch location. Δz: mean (SD) of trajectory height minus reported `plate_z`, in inches, with the trajectory evaluated at the front of the plate or at its midpoint; y^*: where the trajectory meets the reported location.

Source	Pitches	Δz at front (SD)	Δz at midpoint (SD)	Median y^* (ft)
Statcast 2024 (one day)	4,386	+0.003 (0.035)	−0.972 (0.301)	1.41
Statcast 2025 (one day)	8,639	+0.003 (0.035)	−0.978 (0.307)	1.41
Statcast 2026 regular season	714,115	+0.967 (0.299)	+0.003 (0.035)	0.71
Statcast 2026 postseason	9,462	+0.943 (0.308)	+0.003 (0.035)	0.71
Stats API feed 2026, challenged	10,544	—	—	1.42

S3.2. *Reproduction of the ABS rulings.* The ABS zone is a rectangle as wide as the plate (17 in) whose bottom and top are 27% and 53.5% of the batter's certified height, evaluated at the plate's midpoint; a pitch is a strike if any part of the ball (radius 1.45 in) touches it. The ball is round, so the strike region is the rectangle grown by the radius with rounded corners: the signed distance is the radius plus the signed distance of the ball's centre to the rectangle. Growing each edge by the radius with square corners instead, as our pipeline did before the final revision, counts a ball diagonally off a corner as touching it; it relabels 353 called pitches (26 of them challenged) and reproduces 99.71% of the rulings (31 mismatches). Table S5 applies this rule to every challenged pitch. On Statcast's 2026 coordinates as published it reproduces 99.95% of the rulings, and the best constant shift of either zone line is 0.00 in. Projecting the same coordinates from the front to the midpoint a second time lowers every pitch by about 0.95 in, reproduces only 84.8% of the rulings, and makes the best-fitting zone lines sit −1.00 in (bottom) and −0.75 in (top) away from the rule: a double projection of this kind produces an apparent offset between the ABS and Statcast zones. A sweep over the ball radius gives 121 at 1.40 in, 51 at 1.43 in, 5 at 1.45 in, 61 at 1.47 in, 130 at 1.50 in mismatches, sharply minimised at the official 1.45 in (`results/geometry_audit.md`).

TABLE S5

Share of the 10,544 challenge rulings reproduced by the published zone rule.

Coordinates	Rulings reproduced (%)	Mismatches
Statcast 2026 as published, ball radius included	99.95	5
Statcast as published, no ball radius	65.66	3,621
Statcast projected to the midpoint a second time	84.83	1,599
Projected a second time, no ball radius	62.54	3,950
Stats API feed as published (front of plate)	85.00	1,582
Stats API feed projected to the midpoint	99.97	3

Table S6 breaks the reproduction down by edge, batter side, challenger and month. It is uniform: no edge, side, role or month stands out, and there is no drift over the season. The double projection, by contrast, hurt the bottom edge most, where a one-inch vertical shift matters most. The same rule reproduces the rulings in every other feed we use once its front-of-plate coordinates are projected to the midpoint (Table S7), and on the postseason's Statcast coordinates as published it reproduces 100%. The Triple-A and spring-training fits of Section S4.3 therefore use projected feed coordinates, which is correct for those feeds.

TABLE S6
Reproduction of the rulings by group, Statcast 2026 coordinates as published and projected a second time.

		Challenges	As published (%)	Mismatches	Projected twice (%)
Edge	Bottom	3,616	99.89	4	74.8
	Top	1,764	100.00	0	85.6
	Inside (batter's side)	1,844	99.95	1	91.7
	Outside	3,320	100.00	0	91.5
	Pitcher's arm side	2,708	99.96	1	89.7
	Pitcher's glove side	2,456	100.00	0	93.7
Batter	Left-handed	5,203	99.96	2	85.9
	Right-handed	5,341	99.94	3	83.8
Challenger	Batter	4,759	99.94	3	86.2
	Catcher	5,606	99.96	2	83.7
	Pitcher	179	100.00	0	86.0
Month	2026-03	286	100.00	0	86.7
	2026-04	1,640	100.00	0	85.1
	2026-05	1,699	99.94	1	83.7
	2026-06	1,699	99.82	3	84.9
	2026-07	1,643	100.00	0	85.5
	2026-08	1,920	100.00	0	85.3
	2026-09	1,657	99.94	1	84.2

TABLE S7
Reproduction in the Stats API feeds, whose coordinates are at the front of the plate.

Feed sample	Challenges	Front of plate (%)	Projected to midpoint (%)
MLB 2026 regular season	10,557	85.0	99.97
Triple-A 2025	9,341	85.7	99.95
Spring training 2026	1,907	85.3	100.00
MLB 2026 postseason	156	83.3	100.00

Zone lines and height. Statcast's 2026 `sz_top` and `sz_bot` are exactly 53.5% and 27% of a single height on every pitch. Against the whole-inch listed heights of the 662 batters, the lines differ by 0.001 in (SD 0.155) at the top and 0.001 in (SD 0.078) at the bottom; 94.4% of batters are within 0.25 in on both, and the largest difference, 0.267 in, is 0.535×0.5 in: the certified height is measured more finely than the listed one. None of the mismatches below involves a batter whose zone differs from his season median.

S3.3. *The 5 mismatches.* We re-read the live feed (pitch and play description, review flag, challenger, timing) for every mismatch.

- *Boundary ties* (3): Statcast and the feed agree and the pitch is within 0.01 in of the boundary; 1 lie just inside by the rule and were ruled balls, 2 just outside and were ruled strikes. Within 0.5 in of the boundary 0.15% of the 2,698 challenged pitches mismatch; beyond 1 in, 0.02%. With round corners the mismatch rate no longer rises towards the boundary, and the ties show no lean towards either call.
- *Tracking disagreements* (1): Statcast's location differs from the live feed's by inches, and the feed's location agrees with the ruling in every case. Among the 4 challenged pitches where the two sources disagree by more than 0.5 in, the feed is right and Statcast wrong in 1 and the reverse in 0. Six of the eleven mismatches of earlier versions were not tracking disagreements but mis-joined pitches (Section S3.4), whose Statcast location belongs to another pitch; they are now treated as unlocated.
- *Other* (1): a boundary tie in Statcast that the feed places outside the zone.
- *Mislabels*: none. Every recorded call and review flag agrees with the play-by-play text.

The mismatches are spread over the season and over batter and catcher challenges. They are what the mislabel rate $\pi = 0.05\%$ of model (1) absorbs.

S3.4. *Challenges left and the feed's counters.* We reconstruct the challenges each team holds before every pitch: two at the start, one fewer after each failed challenge, plus the extra-inning rule below. The live feed reports, per team and game, the numbers of successful and failed challenges and the challenges remaining (`src/feed_counters.py`, `results/feed_counters.json`). Over the 4,850 team-games of the 2,425 ABS games our counts of successes and failures equal the feed's in 4,849. The exception is game 822717, where one ABS challenge is nested inside a manager's review of the same play and its pitch is ambiguous; we leave it out. The feed's "remaining" equals $\max(0, 2 - \text{failed})$ in 100% of team-games, so it ignores the extra-inning grants; yet teams fail up to 4 times in a game, which is only possible with grants. Our end-of-game count differs from the feed's in 113 team-games, and 0 of these received no grant. No challenge in the data is made with none left (0 cases).

Sample flow. Table S8 follows the called pitches from the feeds to the likelihood and to the policies' opportunity set. A mis-joined pitch is one whose Statcast description contradicts the feed's call code (Statcast reports the final call, so overturned pitches agree); its location belongs to another pitch. A position player pitching is a pitcher with at least 30 plate appearances and fewer than 400 pitches in 2026; MLB's opportunity definition excludes such pitches. Technical unavailability of the system is not flagged in the public data and is not excluded.

TABLE S8
Sample flow, 2026 regular season (2,425 ABS games).

Step	Called pitches	Challenges
Called pitches in the feeds	370,963	10,557
without a Statcast location	−385	
mis-joined to another Statcast pitch	−170	−7
Located called pitches	370,408	10,544
thrown by position players (not opportunities)	−1,143	0
Policies' opportunity set	369,265	
Likelihood sample, batters (challenge held, $\lvert w\rvert < 8$ in)	100,614	4,759
Likelihood sample, fielding side	141,078	5,784

The extra-inning rule. MLB's announcement of the challenge system (Major League Baseball, 2025) gives a team that has used up its challenges one challenge in each extra inning; unused extra-inning challenges do not accumulate. We apply it as: a team at zero receives one challenge at the start of each extra inning, with no stacking. Over the 285 extra innings of the season this grants 151 challenges in 145 team-games, and 210 challenges are made in extra innings. The grant enters the DP and the simulator at the first opportunity of every extra inning, in either half.

Other data notes. 13 of the 10,557 challenges have no Statcast location; they count against the challenge budget but not in the likelihood. The likelihood and the evaluation use called pitches within 8 in of the zone edge; 1 challenges lie farther out (two of them tracking disagreements of Section S3.3).

S4. The perception model.

S4.1. *Specification and estimates.* The criterion is $\kappa(x) = x^\top \beta$ with an intercept, $\log g$, an indicator of one challenge left, the inning divided by nine (capped at ten), and indicators of two strikes and of three balls, separately for batters (on called strikes) and catchers and pitchers (on called balls). In the main specification κ shifts by zone region (bottom, top, inside, outside) and σ_r is multiplied by a region factor, with the bottom edge as reference. The truth is deterministic with mislabel rate π, shared by the two sides. The likelihood (3)–(4) is maximised by L-BFGS followed by a Newton trust region with the exact Hessian from automatic differentiation. Table S9 gives the estimates of the main model and of the model with player random effects.

TABLE S9

Estimates of the main perception model and of the model with player random effects $(\kappa_i, \log \sigma_{r,i})$, bivariate normal per side and integrated by Gauss–Hermite quadrature (5×5 nodes; with 7×7 the log-likelihood changes by 2.7, σ_r by at most 0.4% and the SD of the batters' criterion from 1.18 to 1.26 in). Distances in inches.

	Main model		Random effects	
Parameter	batter	catcher	batter	catcher
κ: intercept	4.65	2.70	5.05	2.83
κ: $\log g$	0.00	0.01	−0.01	0.01
κ: one challenge left	0.46	0.62	0.50	0.68
κ: inning/9	−2.13	−1.69	−2.26	−1.74
κ: two strikes	−2.50	−0.71	−2.58	−0.71
κ: three balls	−0.30	−0.49	−0.29	−0.49
κ: $(\log g + 5)_+$	−0.79	−0.33	−0.76	−0.33
κ: $(\log g + 4)_+$	0.26	0.13	0.25	0.15
κ: $(\log g + 3.2)_+$	−0.11	−0.44	−0.14	−0.42
κ shift, top edge (in)	1.33	0.76	1.18	0.69
κ shift, inside edge (in)	1.03	0.24	0.95	0.22
κ shift, outside edge (in)	1.01	0.09	0.84	0.06
σ_r, bottom edge (in)	2.88	1.82	2.84	1.82
σ_r multiplier, top edge	1.04	1.27	1.00	1.25
σ_r multiplier, inside edge	1.11	1.01	1.10	1.00
σ_r multiplier, outside edge	1.03	0.91	0.98	0.89
Mislabel rate π (%)	0.05		0.05	
SD of player κ_i (in)	—	—	1.18	0.37
SD of player $\log \sigma_{r,i}$	—	—	0.14	0.10
Correlation of player effects	—	—	0.55	0.39

S4.2. *All variants.* Table S10 extends Table 1 with the mislabel rate, BIC relative to the main model, and the wins lost under the pooled DP (no score in the state) next to the score-aware DP of the headline; every row is evaluated with all policies on the same $R = 20$ simulated draws. The smooth-truth rows add, per edge or region, an offset and Gaussian measurement noise to the truth; on the corrected geometry they collapse to the deterministic model (Section 3.1). Without a mislabel rate ("by edge, no mislabel") the measurement noise inflates to absorb the few tracking disagreements of Section S3.3, and the fit is far worse. Heavy-tailed and logistic noise are rejected at equal parameter count. Among the homogeneous probit models that fit at least as well as the Gaussian baseline, wins lost range over 0.63–0.86; the rejected ones give 0.85–1.19. Across the models whose BIC is at least as good as the main model's (lapse and player-effects models included) they range over 0.69–0.92.

TABLE S10

All perception and truth variants. $\Delta\ell$: log-likelihood relative to the Gaussian baseline; ΔBIC: relative to the main model (negative is better); wins lost per team-season under the score-aware and the pooled DP.

Model	k	$\Delta\ell$	ΔBIC	π (%)	σ_r (in)		Wins lost	
					batter	catcher	score DP	pooled DP
Smooth truth by edge, no mislabel	18	−193.0	+912	—	3.04	1.87	0.64	0.54
Smooth truth by region + mislabel	23	+15.8	+556	0.03	3.04	1.87	0.65	0.55
Baseline: deterministic truth + mislabel	15	+0.0	+489	0.05	3.04	1.87	0.65	0.55
Logistic ε	15	−154.2	+797	0.05	2.75	1.74	0.89	0.77
t_{10} ε	15	−106.7	+702	0.05	2.82	1.76	0.85	0.74
t_5 ε	15	−357.9	+1205	0.05	2.74	1.73	1.00	0.89
t_3 ε	15	−852.0	+2193	0.05	2.97	1.88	1.19	1.07
σ_r by edge	17	+63.1	+387	0.05	3.16	1.97	0.66	0.55
σ_r by region	21	+173.1	+217	0.05	3.39	1.94	0.69	0.58
σ_r linear in $\lvert w\rvert$	17	+6.3	+501	0.05	3.21	1.88	0.63	0.52
σ_r by region + slope	23	+183.2	+222	0.05	3.64	1.97	0.64	0.54
κ by region	21	+194.0	+175	0.05	2.98	1.87	0.67	0.57
κ and σ_r by region, κ linear in $\log g$	27	+263.8	+110	0.05	2.89	1.82	0.70	0.58
Same + σ_r slope	29	+271.0	+120	0.05	3.05	1.85	0.66	0.54
Same + lapse rate	29	+277.3	+108	0.05	2.69	1.70	0.85	0.74
Same, κ piecewise linear in $\log g$ (main)	33	+356.0	+0	0.05	2.88	1.82	0.71	0.59
Main + low-stake hinge in κ	35	+356.7	+23	0.05	2.88	1.82	0.71	0.59
Main + σ_r varying with $\log g$	35	+360.8	+15	0.05	2.89	1.82	0.74	0.64
Main, logistic ε	33	+179.6	+353	0.05	2.71	1.72	0.92	0.81
Main, t_{10} ε	33	+230.5	+251	0.05	2.76	1.74	0.89	0.78
Main, t_5 ε	33	−31.8	+776	0.05	2.76	1.73	1.04	0.92
Main, t_3 ε	33	−535.4	+1783	0.05	3.08	1.90	1.23	1.10
Main + lapse rate	35	+371.0	−5	0.05	2.68	1.69	0.86	—
Main + player random effects	39	+1031.3	−1276	0.05	2.84	1.82	0.78	0.67
Main + player effects + lapse rate	41	+1047.1	−1283	0.05	2.69	1.69	0.92	0.82

Fit. The main model predicts 4,753 batter challenges against 4,759 observed and 5,767 fielding-side (catcher and pitcher) challenges against 5,784, and the overturn rates of Section 3. On 22 groups of pitches the summed Pearson discrepancies of the challenge counts are 78 (batters) and 107 (catchers), and of the overturn counts among the model's challenges 56 and 21. The groups come from five overlapping partitions (distance, challenges left, inning, stake tercile, zone region), so the same pitches enter several times and the sums have no simple χ^2 reference; they are descriptive, and with 370,408 called pitches they detect small departures from the probit shape. The random-effects model below fits better.

Nonparametric check of the probit shape. Identification rests on the shape of the challenge curve, and the comparison with logistic and t noise only shows that the Gaussian is best within a parametric family. Figure S1 drops the family. For each side and zone region it

plots the isotonic regression of the challenge indicator on w (monotone under any location-family noise law) with a pointwise 95% band from a game bootstrap ($B = 200$), against the fitted probit curve averaged over the same pitches and smoothed in the same way. The largest gap between the two is 8 percentage points, and the probit curve lies inside the band at 59–100% of the grid points. The departures are local except in the upper tail: for batters at the top and inside edges and catchers at the bottom and inside edges the isotonic curve levels off at about 0.6–0.8 for w beyond 4 in, where the probit curve goes to one. A probit read through such a plateau inflates σ_r. The lapse-rate model of Table S10, $\mathbb{P}(\text{challenge} \mid w) = (1-\lambda)\Phi\{(w-\kappa)/\sigma_r\}$ with one λ per side, fits better by 15 log-likelihood points (2 parameters), gives $\lambda = 15\%$ for batters and 12% for catchers and σ_r of 2.68 and 1.69 in, and gives wins lost of 0.86 with the lapses removed and 0.50 with them kept ($R = 20$; Section 6). Before the feed–Statcast join was checked (Section 2) the plateau, and the lapse model's BIC advantage of more than 200 points, came largely from mis-joined pitches, whose location belongs to another pitch and mostly reads as a badly missed call, and from pitches with essentially nothing at stake. These are pointwise bands, not a simultaneous test.

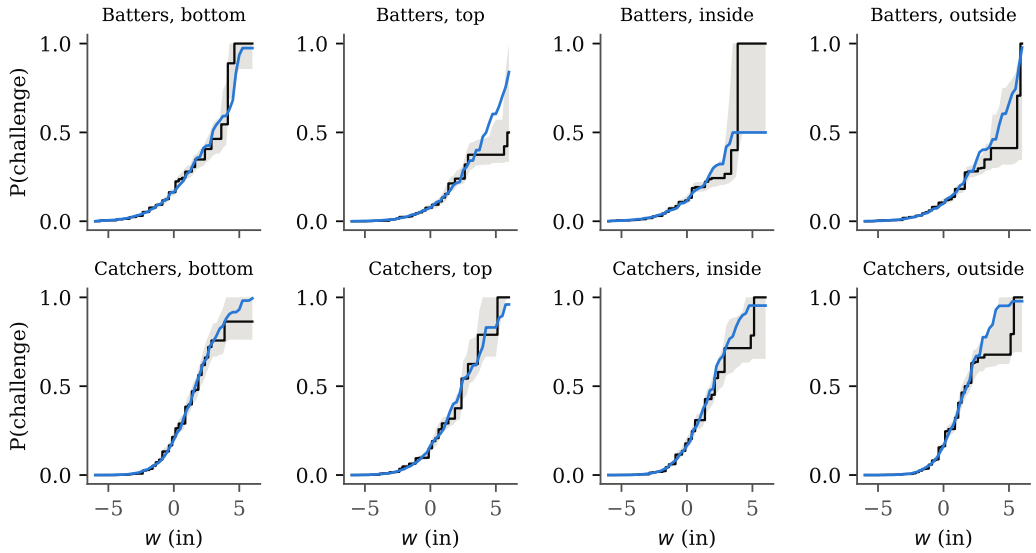


FIG S1. *Challenge probability against the truth margin w (inches, positive: the call was wrong), by side and zone region. Black: isotonic regression of the observed challenges with pointwise 95% game-bootstrap band (grey). Blue: fitted main model (probit), averaged over the same pitches.*

Player heterogeneity. The random effects improve the likelihood by 675 points over the main model and have the best BIC (53,980). They sit mostly in the criterion: the SD of the player criteria is 1.18 in for batters and 0.37 in for catchers, against an SD of $\log \sigma_r$ of 0.14 and 0.10. To evaluate policies we give each player the posterior implied by their own perception noise (empirical-Bayes means), solve the DP with these posteriors and compare with realised play. The wins lost are 0.77 [0.66, 0.92]. The interval comes from a game bootstrap with $B = 150$ in which every replicate redraws the player effects from their empirical-Bayes posteriors without refitting the model; the same bootstrap gives [0.56, 0.82] for the main model, against [0.58, 0.81] with refits. The one-line rule gets 95.4% of the per-player optimum.

S4.3. *Out of sample and other leagues.* We fit the model to March–June (1,279 games) and to July–September (1,146 games) separately. Perception noise is 2.89 in for batters and 1.92 in for catchers in the first half and 2.87 and 1.71 in in the second. Table S11 predicts the challenge counts and overturn rates of the second half from the first-half fit. Overturn rates transfer for both sides and batter challenge counts nearly so, but catchers challenged 22% more than predicted: their criterion moved much more than their perception. Wins lost, computed with the baseline model and the pooled DP (no score in the state) and therefore below the headline, are 0.73 in the first half, 0.48 in the second half with the first-half model and 0.64 with the second-half model.

For Triple-A 2025 and spring training 2026 we use the Stats API feeds, projected to the plate's midpoint (Table S7). The refitted perception noise is 3.22 in [3.07, 3.37] for batters and 2.09 in [2.01, 2.18] for catchers in Triple-A, and 3.78 in [3.39, 4.21] and 2.18 in [2.00, 2.37] in spring training (95% intervals). Transported MLB parameters predict the overturn rates there well but overpredict catcher challenges, again a criterion effect.

TABLE S11
Observed against predicted challenges and overturn rates (observed / predicted). "Fit": the sample the parameters are estimated on.

	Batter		Catcher	
Sample and parameters	Challenges	Overturn (%)	Challenges	Overturn (%)
Mar–Jun, in sample	2,488 / 2,484	47.7 / 49.0	2,835 / 2,828	58.2 / 58.8
Jul–Sep, Mar–Jun fit	2,271 / 2,188	50.2 / 50.4	2,949 / 2,418	57.7 / 58.3
Jul–Sep, Jul–Sep fit	2,271 / 2,269	50.2 / 51.0	2,949 / 2,941	57.7 / 57.8
Triple-A 2025, MLB fit	4,507 / 4,432	45.0 / 50.1	4,832 / 5,446	53.8 / 58.6
Triple-A 2025, refit	4,507 / 4,498	45.0 / 46.3	4,832 / 4,817	53.8 / 54.6
Spring 2026, MLB fit	886 / 915	45.7 / 52.8	1,019 / 1,225	59.5 / 63.1
Spring 2026, refit	886 / 881	45.7 / 46.3	1,019 / 1,007	59.5 / 57.5

Feed coordinates. Rebuilding the 2026 MLB data on the live feed's own coordinates, projected to the midpoint, gives σ_r of 2.90 and 1.84 in in the baseline model and 2.90 and 1.84 in in the main model, against 3.04 / 1.87 and 2.88 / 1.82 in on Statcast; wins lost are 0.62 and 0.64.

S4.4. *Identification and known-truth recovery.* Proposition 1 is checked symbolically (Section S2, checks D1–D4). For the known-truth experiment we simulate challenges and rulings from the fitted baseline model ($\sigma_r = 3.04$ and 1.87 in, $\pi = 0.05\%$) on the real 2026 opportunities, refit from a start with σ_r multiplied by 1.5, and repeat. Table S12 reports the results. Under correct specification the bias is a fraction of a percent. Tracking noise of 0.3 in barely moves σ_r, in line with Proposition 1(c), but it raises the fitted mislabel rate, which absorbs the induced disagreements near the boundary. Heavy tails fitted as Gaussian bias σ_r up by about a quarter, but the data reject heavy tails (Table S10). Ignoring player heterogeneity (simulated with SDs of 1.18 in in κ and 0.14 in $\log \sigma_r$ for batters) also biases σ_r up, which lowers the estimated value of challenging. On the data the bias is smaller: the main model's batter σ_r is 1.4% above the random-effects model's. The direct comparison is the wins lost: 0.77 under the per-player model against 0.69 under the homogeneous one.

TABLE S12
Known-truth recovery of perception noise on the real 2026 opportunities.

Data-generating process	Converged	Bias of $\hat{\sigma}_r$ (%)		SD of $\hat{\sigma}_r$ (in)		Mean $\hat{\pi}$ (%)
		batter	catcher	batter	catcher	
Correctly specified	19/20	+0.3	+0.1	0.03	0.02	0.04
0.3 in tracking noise	10/10	+0.7	+1.3	0.04	0.03	6.18
t_5 perception noise	10/10	+25.4	+22.9	0.07	0.02	0.04
Player random effects	10/10	+12.8	+4.1	0.04	0.02	0.06

S4.5. *Wins lost against perception noise.* The value of every policy falls as perception gets worse, the optimum's fastest. Figure S2 and Table S13 re-solve and re-evaluate the model with σ_r multiplied by a common factor on both sides, holding realised play fixed at 3.34 wins per team-season. Wins lost fall to 0.25 when σ_r is 20% larger than estimated (the break-even of Section 6, by linear interpolation) and to zero near a factor of 1.3. The right panel is the profile log-likelihood of each side's σ_r in the baseline model on the same axis: at the break-even it is 74 points below its maximum for batters and 134 for catchers, and a 10% error already costs 21 and 38.

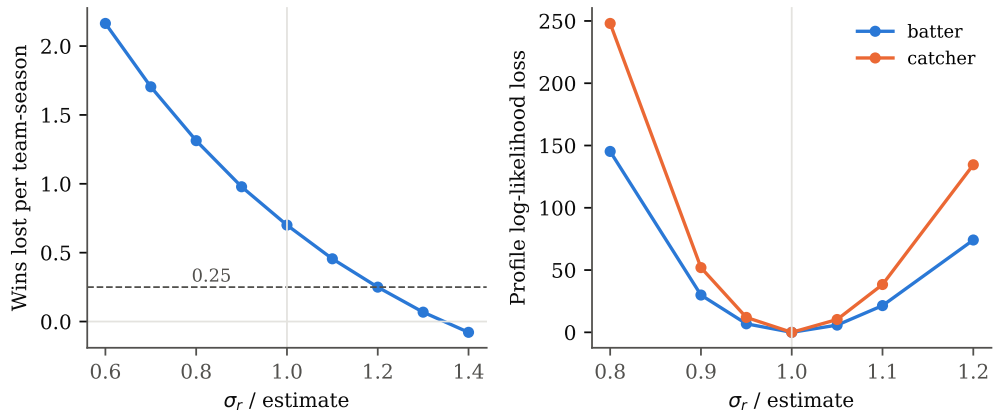


FIG S2. *Left: wins lost per team-season (score-aware optimum minus realised play) when perception noise is a multiple of its estimate; dashed line at* 0.25. *Right: profile log-likelihood loss of σ_r for each side, baseline model.*

Belief maps. Under the calibrated posterior the challenge threshold implied by the fitted criterion is on average $p = 0.34$ for batters and 0.33 for catchers; under a flat-prior belief map it would be 0.89 and 0.81. Neither map makes the implied thresholds respond to leverage as the cutoff (6) requires (Section 6).

TABLE S13

Optimum and wins lost per team-season when σ_r is a multiple of its estimate, and the profile log-likelihood loss at the multiples on the profile grid.

σ_r / estimate	Optimum	Wins lost		Profile $-\Delta\ell$ (baseline model)	
		score DP	pooled DP	batter	catcher
0.6	5.50	+2.16	+2.02	—	—
0.7	5.04	+1.70	+1.58	—	—
0.8	4.65	+1.31	+1.19	145.2	248.0
0.9	4.32	+0.98	+0.86	29.9	52.0
1.0	4.04	+0.70	+0.59	0.0	0.0
1.1	3.79	+0.46	+0.34	21.5	38.4
1.2	3.59	+0.25	+0.14	74.1	134.5
1.3	3.41	+0.07	−0.03	—	—
1.4	3.26	−0.08	−0.19	—	—

S5. Policy evaluation.

S5.1. *Simulated worlds.* Each simulated world keeps every real 2026 called pitch, its stake g, its known truth W and the real call stream. It draws the ruling with mislabel rate π, the signal $s = w + \varepsilon$ from the fitted perception model, and the posterior $p = \mathbb{P}(Y = 1 \mid s)$ that the ruling overturns the call from a weighted isotonic regression of simulated truth on the signal (Barlow et al., 1972). The isotonic map is fitted on an auxiliary, independent draw, so that the truth that scores a policy never enters its decision. Fitting it on the scoring draw instead changed the optimum by -0.004 and the one-line rule by -0.005 wins per team-season (average over $R = 6$ draws). Every policy runs on the same draws, holding the challenges it would hold, with the extra-inning grant of Section S3.4. Table S14 splits the value of every policy by side.

TABLE S14

Wins per team-season by side and challenges per team-game (95% game-bootstrap interval); point estimates of Table 2.

Policy	Wins, batters	Wins, fielding side	Challenges per team-game
Optimum (score-aware DP)	1.83	2.20	2.71 [2.67, 2.75]
Runs-objective rule	1.78	2.17	2.98 [2.94, 3.02]
Pooled DP	1.78	2.14	2.68 [2.65, 2.73]
One-line rule, $\delta = 0.008$	1.73	2.12	2.21 [2.15, 2.27]
One-line rule, $\delta = 0.01$	1.67	2.10	1.90 [1.84, 1.96]
48-cell card, published	1.69	2.11	2.59 [2.54, 2.64]
48-cell card, refit	1.67	2.09	2.52 [2.47, 2.56]
Observed 2026 (realised)	1.44	1.90	2.17 [2.13, 2.22]
Fitted behavioural rule	1.43	1.88	2.15 [2.12, 2.19]
Fixed 0.2-run rule	0.96	1.51	1.02 [0.99, 1.08]

S5.2. *Competitor rules.* *Runs-objective rule* (Namuduri, 2026): challenge iff $p\,\mathrm{RV} > (1 - p)C(t, c)$, where RV is the run value of the flip and C the run value of a challenge at the start of the half-inning, from the same DP solved with runs as the objective. His code indexes C by incorrect challenges spent, which maps to our *c. 48-cell card* (Arora, 2026): a threshold on p for each inning band (1–3, 4–6, 7–8, 9+), tercile of g over all opportunities,

count class (PA-ending iff three balls or two strikes) and challenges held. "Published" uses his 2026 thresholds with our terciles of g; "refit" fills each cell with the median break-even $D/(g+D)$ of our DP over the opportunities in it, a reproduction of his fitting procedure on our model; a card whose cells were chosen to maximise value could do better. Our tercile medians of g are close to his. *Savant's form, tuned*: $p \geq c/(c+\mathrm{RV})$ with c on a grid of run costs, the best chosen at the point estimate (0.08 runs). *Fixed 0.2-run rule* (Baseball Savant, 2026a; Tango, 2026a): challenge iff $p \geq 0.2/(0.2+\mathrm{RV})$. The league computes RV from an RE288 run-expectancy table; we compute it from our own run-expectancy engine, whose run swings are slightly smaller on average, so our version of the rule is marginally more conservative. The ranking does not depend on this: an independent recomputation on Statcast's own run- and win-expectancy changes gives 2.03 wins per team-season for the rule against 3.18 for observed play (ratio 0.64) and 3.71 for the one-line rule. All competitors and the one-line rule are cutoff rules in p (Section S2).

S5.3. *Causal re-simulation.* The path-sum value of a policy is the sum of g over its successful challenges, each g read off the recorded game. After the first flip the recorded continuation is no longer the game that would have been played, which is Arora's objection. For each team-game and policy we find the first divergence between the policy's game and the recorded game: the first called pitch whose call exactly one of them flips (0.76 branch points per team-game for the optimum, 48% of them recorded overturns the policy would have skipped). Before it the two games coincide, so the path sum is exact up to there. At that pitch three worlds are simulated with common random numbers:

A: the call is flipped and the game continues, the team following the policy with the challenges it holds;
A′: the original call stands and the team keeps following the policy;
B: the original call stands and the team never challenges again.

In all three the opponent follows its fitted behavioural rule $s > \kappa(x)$ with its recorded remaining challenges. If the policy flips a call the team left standing, A is the policy's world and A′ the recorded one; if the team flipped a call the policy leaves standing, A (the flipped call) is the recorded world and A′ the policy's, and every correction below enters with the opposite sign. Up to the previous version the branch was the policy's first success, also when the team had flipped the same call, where the recorded continuation is already the flip world. The path-sum value of the team-game is $g_{\mathrm{branch}} + L_{\mathrm{rec}}$, with L_{rec} the team's later gains on the recorded continuation; the causal value is $g_{\mathrm{branch}} + \mathbb{E}[L \mid \mathrm{flip}]$. Two estimators of the correction:

$$\text{RB:}\quad \text{causal} = \text{path} + \mathbb{E}[(L_A - O_A) - (L_{A'} - O_{A'})],$$
$$\text{win count:}\quad \text{causal} = \text{path} + \mathbb{E}[\text{win}_A - \text{win}_{A'}] - g_{\mathrm{branch}},$$

where L (O) sums g over the team's (the opponent's) later successful flips at the simulated state. If the WP engine is the simulator's value function, the team's win indicator is a martingale whose jump at each later flip is $+g$ for the team's flips and $-g$ for the opponent's, so $\mathbb{E}[\text{win}_A - \text{win}_{A'}] = g_{\mathrm{branch}} + \mathbb{E}[(L_A - O_A) - (L_{A'} - O_{A'})]$ and the two estimators agree. The opponent term matters because the flip changes the opponent's later opportunities too; dropping it (the team's own gains only) changes the optimum's RB value by 0.001 wins per team-season; RB is the low-variance version (the decomposition is checked in Section S2). Differencing against A′ cancels any level bias of the simulator's opportunity stream.

Simulator. Pitch events (called pitch, swinging strike or foul tip, foul, ball in play, hit by pitch, automatic ball) by count, outs and bases from 2026 Statcast; called pitches drawn from the 2026 called-pitch pool of the same count, each with its location, truth, signal, posterior and the fitted criterion of that draw; in-play outcomes (outs, bases and runs) by count, outs and bases; baserunning between pitches (steals, wild pitches, passed balls and pickoffs, on about 1.0% of pitches); the runner on second in extra innings; the extra-inning challenge grant; walk-off rules. We use $K = 1000$ play-outs per world and $R = 10$ independent draws of the perception signals.

Checks. In world B the simulator's win rate minus the WP engine's is between -0.012 and -0.011 across policies, with calibration slope 1.000. The offset is the team giving up its challenges while the opponent keeps its own, and it is the same at home and away. Its no-flip continuation produces slightly more later value than the recorded games (1.19 against 1.21 wins per team-season for the optimum; difference -0.03 [-0.09, 0.04]), which the A$-$A$'$ difference cancels.

TABLE S15

Path-sum and causal values, wins per team-season, with 95% intervals for the differences from 2,000 game resamples of the per-game outputs at fixed fitted parameters.

Policy	Path sum	RB	Win count	RB − path	Win count − RB
Optimum (score-aware DP)	4.04	4.04	4.01	+0.000 [−0.001, 0.002]	−0.03 [−0.05, −0.01]
Runs-objective rule	3.95	3.95	3.93	+0.002 [0.000, 0.003]	−0.02 [−0.04, 0.00]
Pooled DP	3.93	3.93	3.91	+0.004 [0.002, 0.005]	−0.02 [−0.04, −0.01]
One-line rule, $\delta = 0.008$	3.86	3.86	3.83	+0.001 [−0.001, 0.002]	−0.02 [−0.04, −0.01]
One-line rule, $\delta = 0.01$	3.76	3.76	3.74	+0.001 [−0.001, 0.003]	−0.02 [−0.04, −0.01]
48-cell card, published	3.82	3.82	3.82	+0.002 [0.000, 0.004]	−0.00 [−0.02, 0.01]
48-cell card, refit	3.78	3.78	3.76	+0.001 [−0.001, 0.003]	−0.02 [−0.04, 0.00]
Fitted behavioural rule	3.31	3.31	3.30	+0.001 [0.000, 0.003]	−0.01 [−0.02, 0.01]
Fixed 0.2-run rule	2.47	2.47	2.46	+0.001 [0.000, 0.003]	−0.01 [−0.03, 0.01]

Results. Table S15: the RB causal value is within a few thousandths of a win of the path sum for every policy. The win-count estimator differs by -0.030 [-0.048, -0.013] wins for the optimum, which puts the wins lost at 0.67 [0.56, 0.78] on that measure; averaged over the 9 policies the difference is -0.018. Section S5.4 explains the remaining difference.

S5.4. *Simulator against WP engine.* The two estimators agree exactly when the simulator's value function is the WP engine. They differ, to first order, by the sum over branch flips of the difference between the simulator's swing in win probability and the engine's g. To measure that difference without Monte Carlo error we write each simulator as an exact pitch-level Markov chain and compute its win probabilities and its g at every branch flip of the optimum (105 per team-season, mean engine g 0.0201). The simulator itself reproduces its chain: from 400 branch states and 2,000 play-outs each, the simulated flip swing minus the chain's is +0.0002 (SE 0.0005), and the simulated home win rate minus the engine's is +0.0002.

Table S16 adds the simulator's ingredients one at a time. A simulator whose in-play outcomes ignore the count and that has no baserunning between pitches moves win probability by only 96.7% of g. That shortfall predicts a win-count minus RB difference of -0.075 wins per team-season for the optimum, and the difference measured with that simulator is -0.015 [-0.059, 0.033]: the shortfall explains the whole gap, with no free parameter. (In the headline run, the simpler simulator's win-count estimate fell short of the g-based accounting by 0.02 wins.) Batters ahead in the count do more damage on contact, so a count-blind in-play

distribution compresses the value of turning a strike into a ball, and the extra pitches a ball buys also give runners more chances to advance. With both ingredients the simulator reaches 99.2% of g; the predicted difference is -0.009 and the one measured in this decomposition run -0.030 [-0.048, -0.013] (-0.030 in the headline run). Table S17 shows that the shortfall of the simpler simulator was largest for flips early in the count.

The path sum is priced in engine g, and the engine is the object calibrated to real outcomes: its distribution of plate-appearance outcomes is the empirical one from every 2026 pitch, and it predicts game outcomes as well as Statcast's own win expectancy (Section S5.8). A residual mismatch between simulator and engine biases the win-count estimate but not the path sum or the RB estimate, which is why the headline uses the path sum.

Home field. Neither the engine nor the simulator has a home-field term, so it cannot separate them. Estimating the engine with separate plate-appearance distributions for home and away batters gives the home team a win probability of 0.527 at the first pitch, changes g at the optimum's branch flips by about -1.1% and changes the wins lost by -0.008.

TABLE S16

Simulator against WP engine at the optimum's branch flips: the simulator's swing in win probability as a share of the engine's g (exact chain), and the win-count minus RB difference that the mismatch predicts, wins per team-season. Each row adds one ingredient to the row above.

Simulator	g / engine g (%)	Predicted win count − RB
Events by count; in-play outcome by the plate appearance's starting outs and bases	96.7	−0.075
+ in-play outcome by count, outs and bases	—	−0.041
+ pitch events by outs and bases	—	−0.061
+ baserunning between pitches (simulator used)	99.2	−0.009
WP engine	100	0

TABLE S17

Simulator swing relative to engine g, minus one, by the count of the branch pitch (optimum).

Count	n	g / engine $g-1$ (%)			
		first simulator	+ contact by count	+ events by outs, bases	+ baserunning
0-0	2,975	−10.4	+1.8	−0.1	+1.6
0-1	1,058	−8.1	−0.1	−0.5	+1.4
0-2	600	−1.1	−4.7	−4.6	−2.2
1-0	1,358	−10.1	−3.5	−6.6	−4.8
1-1	1,111	−8.4	+0.3	−0.4	+1.3
1-2	935	−1.8	−2.8	−3.3	−0.9
2-0	740	+4.5	+2.7	−1.4	+1.7
2-1	826	−3.3	−0.6	−1.0	+1.4
2-2	1,114	−2.2	−3.4	−3.6	−1.1
3-0	348	−1.1	−6.6	−6.8	−3.5
3-1	555	−1.2	−3.2	−3.2	+0.1
3-2	969	−1.3	−2.8	−3.4	−0.5

S5.5. *Behavioural rule against realised play.* The fitted behavioural rule $s > \kappa(x)$ of the main model, simulated holding the challenges it would hold, earns 3.31 [3.16, 3.45] wins per team-season against 3.34 realised. Table S18 compares the two by stake quintile, with the model's expectation taken pitch by pitch at the challenges actually held: the expected value of the teams' challenges is 3.32 wins per team-season against 3.34 realised, and in the top

fifth of stakes the teams made 30 more challenges than the model expects and gained 0.02 more wins.

With a criterion linear in $\log g$, the previous main model, the simulated rule earned 3.22 wins and the excess in the top quintile was large: the real response to the stake is steeper at the top than a linear criterion allows. The main model's piecewise-linear criterion (hinges at $\log g = -5.0$, -4.0 and -3.2, about the 40th, 80th and 95th percentiles) improves the log-likelihood by 92 points with 6 more parameters (BIC better by 110), raises the simulated rule to 3.31 and leaves the decision noise practically unchanged (2.88 in for batters and 1.82 in for catchers, against the linear model's values), so the optimum hardly moves (4.04 against 4.03 wins on the same $R = 10$ draws). The comparison of the optimum with realised play uses the behavioural model only through the decision noise it reveals.

TABLE S18

Observed against model-expected challenges, overturns and value by stake quintile (main model, expectation at the challenges actually held). Value: sum of g over overturned challenges, wins per team-season.

Side	Stake quintile	Mean g	Challenges		Overturned		Value obs. − exp.
			obs.	exp.	obs.	exp.	(wins/team-season)
Batters	1	0.0015	827	820	416	440	−0.000
	2	0.0050	579	618	312	338	−0.005
	3	0.0080	597	599	316	332	−0.005
	4	0.0130	996	953	527	497	+0.012
	5	0.0335	1,760	1,763	756	765	−0.012
Fielding side	1	0.0015	1,009	1,022	551	590	−0.001
	2	0.0050	921	874	587	570	+0.003
	3	0.0081	861	878	558	568	−0.003
	4	0.0131	1,092	1,125	672	668	+0.003
	5	0.0360	1,901	1,868	984	965	+0.030

S5.6. *Thresholds by stake.* Table S19 gives, by stake quintile, the confidence threshold the teams' fitted criterion implies through the calibrated belief map and the DP benchmark's threshold $D/(g + D)$ at the recorded state, and the same in inches of signal: the teams' criterion κ and the signal at which the belief reaches the one-line rule's threshold $0.008/(g + 0.008)$. Teams' thresholds fall far less with the stake than the benchmark's.

S5.7. *The flat-optimum bound on the calibrated pools.* To evaluate Theorem 4 exactly we need a model in which its assumptions hold. We use the score-aware calibrated DP restricted to regulation: the exogenous state is the half-inning and the home score difference at its start; the number of opportunities in a half-inning follows its empirical law in that state (as in the DP) and their (g, p) are drawn i.i.d. from the 2026 pool of that state; runs between half-innings come from their empirical law; and the game ends after the top of the ninth if the home team leads and after the ninth otherwise. In this model the MDP is exact, so identity (i) must hold to rounding. Bound (ii) needs the density bound M; we estimate it as the largest histogram density of p (bin width 0.025) within quintiles of g, and the "local" approximation replaces it by the density at the midpoint of each interval. Table S20 gives the results ($R = 3$ draws). Identity (i) holds to the printed precision; the best constant $\delta = 0.008$ loses 6.0%; the local approximation is close (6.4%); and (ii) with the histogram heights gives 45.5. Because the calibrated posterior is a step function of the signal (it is fitted on bins of 0.1 in), its law has atoms and no density, so that number is a smoothed approximation, not a certified bound;

TABLE S19
Revealed and benchmark thresholds by stake quintile (eligible pitches, main model, R draws of the belief map). Signal thresholds are averages over the pitches of the quintile.

Side	Stake quintile	Median g	Confidence threshold p^*		Signal threshold (in)	
			teams	DP benchmark	teams κ	one-line rule
Batters	1	0.0014	0.46	0.65	4.1	9.3
	2	0.0052	0.48	0.55	4.5	6.7
	3	0.0081	0.47	0.48	4.5	5.1
	4	0.0136	0.44	0.34	3.7	3.0
	5	0.0281	0.35	0.19	2.4	0.3
Fielding side	1	0.0013	0.45	0.65	1.8	4.9
	2	0.0051	0.50	0.55	2.1	2.9
	3	0.0082	0.50	0.47	2.1	2.1
	4	0.0138	0.44	0.34	1.7	1.2
	5	0.0300	0.35	0.19	1.1	0.0

identity (i) needs no density. On the real game stream the same rule loses 5%. The two problems differ: in the exact model the DP is optimal by construction, while on the real stream it is optimal for the pooled law it was solved on, not for the realised sequence of calls and stakes, and the real stream includes extra innings, where the top-up rule makes the DP and the rule act alike. The rule's share is relative to the benchmark, and the clairvoyant ceiling of Section 4 bounds what the benchmark leaves.

TABLE S20
Theorem 4 on the calibrated pools, home and away sides pooled. Loss: $1 - V^{\delta}/V^{\star}$. Identity gap: largest difference between $V^{\star} - V^{\delta}$ and the sum in (i).

δ	$V^{\delta}/V^{\star}$ (%)	Loss (%)	Local approx. (%)	Bound (ii) (%)	Identity gap
0.004	80.1	19.9	15.9	49.9	0
0.005	87.3	12.7	10.8	36.8	0
0.006	91.5	8.5	7.7	32.7	0
0.007	93.6	6.4	6.3	36.1	0
0.008	94.0	6.0	6.4	45.5	0
0.009	93.3	6.7	7.6	59.0	0
0.010	91.9	8.1	9.3	76.2	0
0.012	87.7	12.3	14.1	118.9	0
0.015	80.3	19.7	22.7	196.4	0
0.020	68.6	31.4	36.3	343.4	0

S5.8. *Bootstrap.* The resampling unit is the game. Each of the $B = 397$ replicates draws the 2,425 games with replacement under new identifiers, refits the truth and perception model, refits the isotonic posterior maps, re-solves every DP, re-selects the one-line rule's δ and re-simulates every policy on $R = 10$ draws. The WP engine, the stakes and the run distributions between half-innings stay fixed; they are estimated from all 2026 pitches, and the joint bootstrap below adds their uncertainty. The intervals are percentiles of the replicates. For the wins lost the basic interval is [0.57, 0.80] against the percentile interval [0.58, 0.81], and the bootstrap SD is 0.057.

Monte Carlo error. Each replicate averages over $R = 10$ simulated draws. At the point estimate the SD of the wins lost across single draws is 0.053 (the 40 draws of the point estimate), so Monte Carlo noise contributes about $0.053/\sqrt{10} = 0.017$ to the bootstrap SD of 0.057, which adds little to the interval width. An earlier run with $R = 2$ draws per replicate and $B = 120$ replicates gave [0.50, 0.75] (SD 0.069). The point estimate averages over $R = 40$ draws, with Monte Carlo standard error 0.008.

Coverage. The study uses the final pipeline and main model. We simulate 50 seasons from the fitted model on the real 2026 call stream. Every called pitch keeps its tracked location, the ruling is drawn with the fitted mislabel rate, and each team challenges by the fitted rule $s > \kappa(x)$ with the challenges it holds in that simulated season. The true gap is the optimum's value minus the expected value of that rule, 0.72 wins per team-season (400 simulated seasons), close to the headline gap because the fitted rule earns about what teams realised. Each season is then analysed as the data are: the model is refitted, the point estimate averages $R = 10$ draws, and the refit bootstrap uses $B = 50$ replicates of $R = 4$ draws, fewer than the $B = 397$ and $R = 10$ of the analysis, for computing time; 35 of the 2,550 refits did not converge and are dropped. The estimand of this study is therefore the gap to the fitted rule in a world where that rule is the truth, not the gap to realised play; what it checks is the interval procedure. The percentile interval covers the true gap in 94% [83.5, 98.7] of the seasons (Clopper–Pearson interval); it misses below in 0 and above in 3. The basic interval covers in 98% and the normal-approximation interval in 98%. Across seasons the point estimate has bias 0.006 and SD 0.044, against a mean bootstrap SD of 0.058.

Sampling error of the WP engine. Each of $B = 100$ replicates resamples the 2026 Statcast games, re-estimates the plate-appearance transition probabilities, the WP and run-expectancy engine and the run distribution between half-innings, recomputes g and RV for every called pitch, and re-evaluates every policy at the point parameters on the same $R = 10$ draws. The wins lost move by $[-0.010, 0.045]$ (SD 0.014, 0.25 of the refit-bootstrap SD 0.057). Adding these deviations to the refit replicates, all pairs, gives [0.60, 0.83] against [0.58, 0.81]. The one-line rule's share of the optimum stays within [95.1, 95.9]%.

Joint bootstrap. Each of $B = 100$ replicates resamples the games once and rebuilds everything on that resample: the plate-appearance transitions, the WP and run-expectancy engine, the stakes g and RV of every called pitch, the run laws of the DP (regulation and extra innings), the perception fit, the belief maps and the policies ($R = 4$ draws). The wins lost have the 95% interval [0.61, 0.85] with SD 0.067, 1.17 times the SD of the game bootstrap that holds the engine fixed; the one-line rule's share stays within [94.1, 96.3]%. This replaces the independent-component combination of earlier versions.

Block bootstrap. To allow for dependence between games of the same team and period (catchers and pitchers who recur, umpire crews, the criterion's drift through the season), $B = 100$ replicates resample blocks of a home team's games in a calendar month, refit the model and re-evaluate ($R = 4$). The wins lost have the interval [0.61, 0.80], with SD 0.97 times the game bootstrap's.

Calibration on the selected pitches. The DP prices a challenge with the belief p, so p must be calibrated on the pitches a policy selects, not only on average. Across R draws, the realised stake-weighted success $\sum Wg$ over the pitches the optimum challenges exceeds $\sum pg$ by 1.1% (0.4% for the one-line rule). With calibration cells by side and region only, as in earlier versions, the excess was larger; the remaining excess means the beliefs are slightly underconfident on the selected pitches, which makes the benchmark slightly cautious.

Selection of the one-line rule. Across replicates the re-selected constant was $\delta = 0.008$ in 333, $\delta = 0.007$ in 62, $\delta = 0.009$ in 2, so the share 95.3% [94.4, 96.1] quoted in the main text includes this selection.

The WP engine. The engine is estimated by backward induction from 2026 transitions. Its Brier score for the game outcome over all plate appearances is 0.1585, against 0.1585 for Statcast's own win expectancy (correlation 0.9996). It has no home-field term: home teams won 52.9% of games, and in tied states the engine is 2.4 percentage points below the realised home win rate. Because g is a difference of win probabilities, a level shift of this kind is second order for the stakes; adding a home-field term changes the wins lost by -0.008 (Section S5.4). With game-clustered standard errors, 3 of 48 inning-by-score cells deviate by more than three standard errors.

S6. The 2026 postseason.

S6.1. *Sample and design.* We use the 33 postseason games completed by 17 October 2026 (Table S21); the World Series is not included. The data are built exactly as for the regular season, on Statcast's 2026 coordinates as published, and the zone rule reproduces every postseason ruling; no challenge is made with none left. Every model parameter, the isotonic belief map and the score-aware DP come from the regular season. Postseason data enter only the explicitly labelled postseason refits.

TABLE S21

The 2026 postseason by round, and the regular season for comparison. Challenges are those on pitches with a Statcast location (10,544 of the 10,557 regular-season challenges); the catcher column includes pitchers.

	Games	Called pitches	Challenges	Per team-game	Batter	Catcher, pitcher	Overturned (%)
Wild Card	9	1,315	41	2.28	16	25	58.5
Division Series	16	2,431	80	2.50	35	45	58.8
LCS	8	1,145	35	2.19	9	26	68.6
Postseason	33	4,891	156	2.36	60	96	60.9
Regular season	2,425	370,408	10,544	2.17	4,759	5,785	53.9

S6.2. *Perception transfer.* Table 4 of the main text compares observed with predicted challenges. The challenge log-score skill is $1-\ell/\ell_0$, where ℓ is the mean log score of the challenge decision on eligible pitches and ℓ_0 that of a constant probability equal to the model's mean. With a deterministic truth the overturn rate given a challenge checks only the geometry, and it is reproduced exactly. Refitting both σ_r and the criterion's intercept on the postseason shifts the catchers' criterion by -0.34 [-0.70, 0.03] in, which absorbs their extra challenges (batters: -0.53 [-1.22, 0.17] in), and gives σ_r of 2.40 [1.99, 2.89] and 1.73 [1.48, 2.03] in, whose intervals overlap the regular-season ones.

S6.3. *Learning.* Table S22 applies the full-season regular-season model to each period's opportunities, so observed against predicted isolates behaviour from the opportunity mix. Batters challenge at a steady rate per eligible pitch. Catchers challenged less than the model predicts before the All-Star break (14 July) and more after it, and the postseason continues the second-half rate.

TABLE S22
Challenges per 100 eligible pitches and observed against predicted challenges, full-season model.

Period	Games	Challenges per 100 eligible pitches		Observed / predicted (z)	
		batter	catcher	batter	catcher
Before the All-Star break	1,442	4.66	3.81	2,807 / 2827.7 (-0.4)	3,225 / 3476.2 (-5.1)
After the All-Star break	983	4.83	4.53	1,952 / 1925.0 ($+0.7$)	2,559 / 2291.0 ($+6.7$)
Postseason	33	4.42	5.02	60 / 61.5 (-0.2)	96 / 79.6 ($+2.2$)

S6.4. *Series stakes.* Challenges do not carry over between games. With a coin-flip series model, the probability of winning the series is $P(\text{series} \mid \text{lose}) + L\,P(\text{win game})$, where $L = P(\text{series} \mid \text{win}) - P(\text{series} \mid \text{lose})$ is fixed within a game. Maximising series win probability within a game is therefore exactly maximising game win probability: series stakes are game stakes scaled by the factor $L \leq 1$, which scales g and the continuation value Δ together and leaves the cutoff (6) unchanged. A stake-rational threshold should not move with L. Refitting the criterion on $\log(gL)$ instead of $\log g$ fits no better ($-\log$-likelihood 360.55 against 358.35 at the same parameter count), and the excess challenges in the 14 games with $L \geq 0.5$ are not significant ($z = +1.1$ for batters and $+1.3$ for catchers).

S6.5. *Value.* Table S23 evaluates the policies by path sum with the regular-season belief map and DP ($R = 20$ draws), in game win probability per team-game. The postseason interval is a game bootstrap ($B = 2000$) at fixed regular-season parameters; it omits parameter uncertainty, which is small next to the sampling noise of 33 games. Weighted by series leverage, the loss is 0.0008 [-0.0013, 0.0027] series win probability per team-game.

TABLE S23
Value of challenges in game win probability per team-game (challenges per team-game in parentheses), 95% game-bootstrap interval for the postseason.

Policy	Regular season	Postseason
Optimum (score-aware DP)	0.0248 (2.71)	0.0260 (2.72)
One-line rule, $\delta = 0.008$	0.0237 (2.23)	0.0256 (2.35)
Fitted behavioural rule (simulated)	0.0205 (2.16)	0.0212 (2.06)
Observed (realised)	0.0207 (2.17)	0.0238 (2.36)
Optimum $-$ observed	0.0042	0.0022 [-0.0027, 0.0066]
Share of the optimum captured (%)	83	92 [74, 110]

S7. Cricket's Decision Review System: an exploratory application. This application lacks the device that identifies the MLB model: the ball-tracking truth is not public, so the truth enters as a latent margin, the perception noise is identified only through the shape of the review curve and the share of umpire's calls, and the results below are exploratory. For an economic analysis of DRS decisions see Shivakumar (2018).

S7.1. *Data and review allowances.* We use Cricsheet ball-by-ball data (Rushe, 2026): men's Tests, ODIs and T20 internationals from October 2017 and the IPL from 2018, with T20 internationals and IPL pooled as "T20". A match is in the sample if it has a recorded review, or none but another match of the same competition, event and season between the same teams or at the same city has one (DRS is set per series and host); matches that Cricsheet flags as missing their review records are excluded (Table S24). Keeping only matches with a review would select on the behaviour being modelled. ODIs have more zero-review matches

than a Poisson count of reviews implies, so some of the added ODIs may have been played without DRS; Cricsheet does not record it. Reviews follow the ICC playing conditions: from October 2017 a review that ends in umpire's call is kept; Tests allow two unsuccessful reviews per innings, ODIs and T20 internationals one; from July 2020 the interim regulations add one in every format; the IPL allows one until 2021 and two from 2022. We reconstruct the reviews each side holds from these rules. The data agree with them half-year by half-year: the maximum number of unsuccessful reviews per side and innings equals the allowance in every half-year and competition, and only 2 of 8,440 reviews are taken with none left. An independent recount from the raw files reproduces every count in Table S24.

TABLE S24

DRS data. DRS matches include the zero-review matches of DRS series (Section S7). Batting-side out decisions are every LBW, caught or caught-and-bowled out decision, including those overturned on review.

	Test	ODI	T20
DRS matches	331	623	1,331
Batting-side out decisions	8,683	6,738	12,246
of which LBW	1,497	824	862
Batting reviews	1,822	809	914
overturned / umpire's call / lost	642 / 377 / 803	310 / 146 / 353	321 / 165 / 428
Fielding reviews	2,326	1,092	1,477
overturned / umpire's call / lost	456 / 350 / 1520	263 / 149 / 680	329 / 188 / 960

S7.2. *Model and likelihood.* On the batting side the opportunity set is observed: every out decision could be reviewed. A decision of kind k (LBW or caught) has a latent margin $X \sim N(\mu_k, 1)$ and is overturned on review iff $X > 0$. A failed LBW review is kept (umpire's call) iff the margin is marginal, $-\kappa < X < 0$, and lost iff $X \le -\kappa$; a failed caught review is always lost. The batter sees $S = X + \varepsilon$, $\varepsilon \sim N(0, s^2)$, so s is perception noise in units of the spread of decision margins, and holds the Bayes posterior

$$X \mid S \sim N(m, v), \quad m = \frac{\mu_k s^2 + S}{1 + s^2}, \quad v = \frac{s^2}{1 + s^2}, \qquad p = \mathbb{P}(X > 0 \mid S) = \Phi(m/\sqrt{v}).$$

Since p is increasing in S, reviewing iff $p \ge \tau$ is reviewing iff $S \ge S^*(\tau) = (1 + s^2)\sqrt{v}\,\Phi^{-1}(\tau) - \mu_k s^2$. The revealed threshold τ is estimated per cell of reviews left and wickets down (0–2, 3–5, 6–9). With $q = \mathbb{P}(S \ge S^*)$ and the orthant probability $J(t) = \mathbb{P}(X > t, S \ge S^*) = \Phi_2\{\mu_k - t, (\mu_k - S^*)/(1 + s^2)^{1/2}; (1 + s^2)^{-1/2}\}$, the likelihood of an out decision is

$$\begin{aligned} &\text{not reviewed:} \quad 1 - q, && \text{overturned:} \quad J(0), \\ &\text{umpire's call:} \quad J(-\kappa) - J(0), && \text{lost:} \quad q - J(-\kappa) \ \ (\text{caught: } q - J(0)), \end{aligned}$$

mixed over kinds with the LBW share for overturned decisions, whose kind is not recorded. The "constant-rate" alternative keeps a failed LBW review with probability ρ, independent of X. Every closed form is checked in Section S2.

Identification. Outcome rates identify the review share, success rate and umpire's-call share in every cell, the share of wrong LBW decisions and the band κ (Table S25). Perception noise is harder. As τ moves across cells, the review, success and umpire's-call shares trace one curve whose shape bounds s from below; the umpire's-call channel adds information because it tells how close the reviewed decisions were. Without it (constant-rate model) the

likelihood is flat above $s \approx 0.5$. With it, the fitted s is consistent with the data in Tests and T20 and weakly identified in ODIs. The upper bound is soft even in Tests and T20: in known-truth recovery from the fitted model ($R = 20$ replicates per format) the estimate exceeds 1 in 15–40% of replicates, although the profile-likelihood interval keeps close to its nominal coverage. The intervals are continuous profile-likelihood intervals from multi-start fits.

TABLE S25
DRS selection model, batting side, and known-truth recovery of s.

	Test	ODI	T20
Perception noise $\hat{s}$ (margin SD)	0.47	0.81	0.45
95% profile-likelihood interval	[0.4, 0.8]	[0.4, 2.3]	[0.4, 0.9]
Umpire's-call band $\hat{\kappa}$	0.48	0.41	0.40
LBW out decisions that are wrong (%)	30.3	29.0	27.8
Caught out decisions that are wrong (%)	0.08	0.14	0.02
Constant-rate model: $P(\text{kept} \mid \text{failed})$	0.36	0.31	0.31
Known truth: median $\hat{s}$ (truth = fitted $\hat{s}$)	0.50	0.78	0.64
replicates with $\hat{s} > 1$	4/20	8/20	3/20
coverage of the 95% LR set (%)	90	95	90

S7.3. *Dynamic program and policies.* The win-probability engine for each format is gradient boosting with monotone constraints, without recalibration: an isotonic map fitted on held-out matches and scored on other ones does not improve the Brier score (it slightly worsens it in Tests), so the stakes use the engine's own predictions; Tests use expected match points (win $+\frac{1}{2}$ draw). The stake g of a decision is the change in match value from one more wicket at that state. The DP state is wickets down w and reviews left c within a batting innings. From w the next event is a reviewable out decision (probability a_w), another wicket (b_w) or the end of the innings, with rates estimated from the real innings. With $U(w, c)$ the expected stake recovered by reviews from (w, c),

$$U(w,c) = a_w \mathbb{E} \max\{U(w+1,c),\ p\,(g + U(w,c)) + (1-p)U(w+1,c) - \ell\,\Delta U\} + b_w U(w+1,c),$$

with $\Delta U = U(w+1,c) - U(w+1,c-1)$ and ℓ the posterior probability that the review is lost. It is optimal to review iff $p(g + A) \geq \ell\,\Delta U$, $A = U(w,c) - U(w+1,c)$. Policies are simulated on the real decision streams: the optimum, the best one-line rule $p \geq \delta/(g+\delta)$, teams' fitted revealed rule, and $p \geq 0.5$; the realised value of the real reviews is reported alongside. Uncertainty comes from a match bootstrap with $B = 500$ that refits the selection model, re-solves the DP and re-simulates.

S7.4. *Results.* Table S26 gives the values in match points per 100 team-matches. The one-line rule is within about one percent of the optimum in every format. As in MLB, the primary comparison is the optimum against realised play: in Tests teams lose 0.13 [−0.29, 0.43] match points per 100 team-matches, an interval that contains zero. Against teams' fitted rule simulated on the same draws as the optimum the loss is 0.020. The two differ by the luck of which decisions were actually overturned and by how well the fitted rule reproduces realised play; unlike in MLB (Section 5), the fitted rule's value lies inside the interval of the realised value, so here the two comparisons tell the same story. The optimum reviews more often than teams do, but with $\hat{s} \approx 0.5$ the posteriors are nearly bimodal and the extra

TABLE S26
DRS, batting side: match points per 100 team-matches, 95% match-bootstrap intervals ($B = 500$).

	Test	ODI	T20
Optimum	5.61 [4.96, 6.16]	2.28 [2.00, 2.78]	0.96 [0.85, 1.08]
One-line rule	5.60 ($\delta = 0.001$)	2.25 ($\delta = 0.005$)	0.96 ($\delta = 0.0005$)
share of optimum (%)	99.8 [99.3, 100.0]	98.5 [96.0, 99.9]	99.5 [98.9, 100.1]
Teams' fitted rule (simulated)	5.59	2.26	0.96
Observed (realised)	5.47 [4.83, 6.12]	2.28 [1.99, 2.62]	0.94 [0.82, 1.06]
Review iff $p \geq 0.5$	4.30	1.42	0.75
Lost vs fitted rule	0.020 [0.010, 0.065]	0.018 [−0.008, 0.394]	0.001 [−0.002, 0.008]
Lost vs realised	0.13 [−0.29, 0.43]	0.00 [−0.13, 0.44]	0.02 [−0.04, 0.09]
Reviews per team-match: optimum / rule / observed	4.44 / 1.94 / 2.75	1.02 / 0.59 / 0.65	0.48 / 0.28 / 0.35

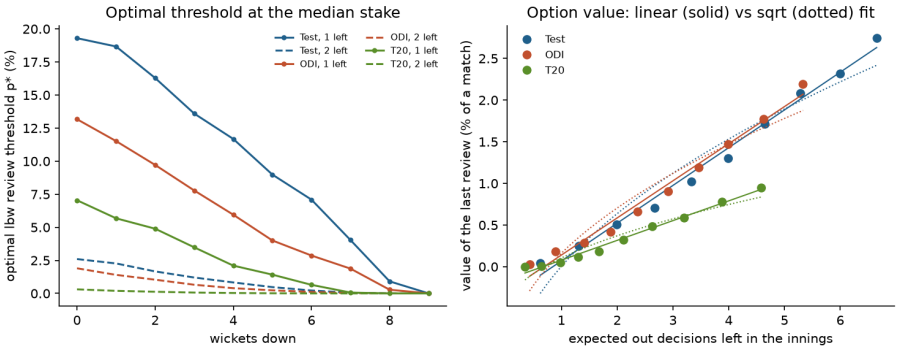


FIG S3. *DRS replication, batting side, by format.*

reviews are of decisions with $p \approx 0$, worth almost nothing either way: there is no counterpart to MLB's loss.

Table S27 gives the optimal LBW threshold (posterior probability that the decision is wrong, in percent) at each number of wickets down. With one review left it falls steadily from the top order to the tail; with two it is near zero throughout. Table S28 fits the value of the last review against the expected out decisions left, N. A linear fit beats $\sqrt{N}$ in every format and the power exponent is well above $1/2$: with five to seven out decisions per innings on average and about 30.3% of LBW out decisions wrong, the last review is worth roughly the expected errors ahead times the stake. Exponents above one cannot come from the stationary model of Theorem 3, where the increments $\Delta_t(1) - \Delta_{t-1}(1) = \varphi(\Delta_{t-1}(1))$ are non-increasing (Lemma S1 with Theorem 2(b)) and growth is at most linear; they reflect stakes and decision rates that change with the wickets down. What carries over is only that the square-root law needs many low-probability opportunities per retained challenge, which a batting innings does not have.

S7.5. *Sensitivity and limitations.* Table S29 varies perception noise, the umpire's-call model, the WP engine and the stakes. Losses become material only if perception is poor ($s = 2$), which the likelihood rejects in Tests and T20 but not in ODIs, where the no-loss conclusion rests on the point estimate of s. The one-line rule stays within a few percent of the optimum in every variant. On the fielding side only reviews are observed; not-out appeals that were not reviewed leave no trace, so the opportunity set, the optimum and the loss cannot be estimated. Descriptively, fielding reviews succeed in 20, 24 and 22% of cases (Test, ODI, T20) and end in umpire's call in 15, 14 and 13%.

Limitations. The WP engines are league-average, with no team strength or conditions. The Gaussian margin model is parametric, and its shape beyond the observed thresholds is as-

TABLE S27
Optimal LBW review threshold (%) by wickets down and reviews left.

Wickets down	0	1	2	3	4	5	6	7	8	9
Test, 1 left	19.3	18.7	16.3	13.6	11.7	9.0	7.1	4.0	0.9	0.0
Test, 2 left	2.6	2.3	1.7	1.2	0.8	0.5	0.2	0.0	0.0	0.0
ODI, 1 left	13.2	11.5	9.7	7.8	6.0	4.0	2.9	1.9	0.3	0.0
ODI, 2 left	1.9	1.4	1.0	0.7	0.4	0.2	0.1	0.0	0.0	0.0
T20, 1 left	7.0	5.7	4.9	3.5	2.1	1.4	0.6	0.1	0.0	0.0
T20, 2 left	0.3	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0

TABLE S28
Value of the last review against the out decisions left.

	Test	ODI	T20
R^2 of Δ_1 on $\sqrt{N}$	0.940	0.923	0.935
R^2 of Δ_1 on N	0.989	0.985	0.989
Power exponent [95% CI]	1.66 [1.51, 1.80]	1.68 [1.49, 1.86]	1.85 [1.57, 2.13]
Expected out decisions left at the start, N	6.65	5.33	4.58

TABLE S29
DRS sensitivity, batting side: optimum / lost vs fitted rule / one-line rule's share of the optimum (%), match points per 100 team-matches. $-\Delta\ell$: profile log-likelihood loss of s.

	Test	ODI	T20
Main (band model, $\hat{s}$)	5.61 / 0.020 / 99.8	2.28 / 0.018 / 98.5	0.96 / 0.001 / 99.5
Upper end of the s set	5.60 / 0.053 / 98.7 ($s = 0.84$)	2.94 / 0.724 / 93.9 ($s = 2.28$)	0.97 / 0.016 / 99.2 ($s = 0.86$)
$s = 2$	6.66 / 1.174 / 95.4 ($-\Delta\ell = 5.21$)	2.78 / 0.549 / 93.8 ($-\Delta\ell = 1.57$)	1.38 / 0.452 / 93.8 ($-\Delta\ell = 4.46$)
Constant-rate umpire's call	5.39 / 0.032 / 99.7	2.33 / 0.023 / 99.5	0.96 / 0.000 / 99.8
Recalibrated WP engine	5.70 / 0.022 / 99.8	2.32 / 0.017 / 98.5	1.01 / 0.002 / 99.4
Stakes $\times 1.5$	8.41 / 0.030 / 99.9	3.42 / 0.027 / 98.5	1.44 / 0.002 / 99.5
Stakes $\times 0.67$	3.76 / 0.013 / 99.9	1.53 / 0.012 / 98.3	0.64 / 0.001 / 99.6

sumed. The DP pools stakes by wickets down, so overs left and declarations enter only through the empirical rates. The counterfactual keeps the real decision stream fixed: a batter reprieved in simulation creates no new decisions, the same approximation as the MLB path sum, and there is no causal re-simulation for DRS; solving the DP without the extra opportunity after a successful review, so that DP and evaluation share one process, moves the optimum by at most a few thousandths of a match point. The best δ chosen on one half of the matches by date and scored on the other keeps 98–100% of the optimum.

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